b) Show that if S and T are two sets each with m elements, where m is a positive integer, then there is a one-to-one correspondence between S and T.

Soln: WLOG, Let $S=\{a_1, a_2, \ldots, a_m\}$ and $T=\{b_1, b_2, \ldots, b_m\}$.

Let $f: S \to T$, where $f(a_i)=b_i$ where $1 \leq i \leq m$, $i \in N$.

It's easy to see that $\forall y \in T, \exists x \in S$ s.t. $f(x)=y$.

By Defn. 7, f is onto. By Prob 74, f is one-to-one.

By Defn. 8, f is a one-to-one correspondence. □

82) Show that a set S is infinite if and only if there is a proper subset A of S such that there is a one-to-one correspondence between A and S.

Soln: **If portion:** We shall give a proof by contrapositive.

Let S be a finite set and $|S|=n$. WLOG, Let $S=\{a_1, a_2, a_3, \ldots, a_n\}$

By Defn, any arbitrary proper subset A of S has atleast one element x s.t. $x \in S$ and $x \notin A$. $\therefore |A| < |S|$.

Let $f: A \to S$ be a one-to-one correspondence.

By Defn 8, f is both one-to-one and onto.

$\therefore |A| < |S|$, $\exists y, z \in S$ s.t. $y \neq z$ and $f(b_i)=y$ and $f(b_i)=z$ for some $b_i \in A$. This is a contradiction to the defn. of function in Defn. 1.

$\therefore$ y and z must exist due to the defn. of onto function in Defn. 7.

$\therefore$ There does not exist a proper subset A s.t. A has a one-to-one correspondence with S.

Only if portion: We shall give a proof by contraposition.

Let, there does not exist a proper subset A of S where A has a one-to-one correspondence between A and S.